

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 5

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1. TITLE

Develop a Dynamic Webpage Using JavaScript DOM Manipulation

2. AIM

To design and develop a web page that uses JavaScript DOM manipulation to dynamically modify element content, change CSS styling properties, alter HTML attributes, and create and append new DOM elements at runtime.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and use `document.getElementById()` to select elements in the DOM tree.
 - To update element text and inner HTML content using `innerHTML`.
 - To dynamically modify CSS styles using `element.style` properties.
 - To change element attributes such as `image src` and `width` dynamically.
 - To create and append new elements dynamically using `document.createElement()` and `appendChild()`.
 - To handle user interaction using `onclick` event handlers.
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4. THEORY

DOM Element Selection — `document.getElementById()` retrieves a unique DOM element node by its ID attribute, enabling direct programmatic manipulation.

Content Modification — the `innerHTML` property gets or sets the HTML markup and visible text contained within an element, allowing dynamic rewriting of content.

Dynamic Styling — the style property gives direct access to an element's inline CSS rules (such as color, backgroundColor, and fontSize), overriding static stylesheet declarations.

Attribute Manipulation — element attributes such as src and width can be read and reassigned dynamically at runtime to swap resources or adjust dimensions.

Dynamic Node Creation — document.createElement() instantiates a detached node in memory, and appendChild() attaches it to a parent node within the document body.

5. ALGORITHM / PROCEDURE

- Create an HTML file named index.html.
 - In index.html, add an <h2> with id="heading", a <p> with id="message", and an with id="image".
 - Add four buttons with onclick event listeners: changeContent(), changeStyle(), changeAttribute(), and addElement().
 - In the <script> section, define changeContent() to update the innerHTML of heading to "Welcome Students" and message to "Content changed using the DOM."
 - Define changeStyle() to set heading text color to "blue", background color to "lightgray", and message font size to "20px".
 - Define changeAttribute() to change the image src attribute to "new.jpg" and width to 300.
 - Define addElement() to create a new <p> element with innerHTML "This paragraph was created using JavaScript." and append it to document.body.
 - Open index.html in a web browser, test all button actions, and record expected vs. actual results.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Manipulation</title>
</head>
<body>

  <h2 id="heading">Student Information</h2>

  <p id="message">Click the button to change this content.</p>
```

```

```

```
<br><br>
```

```
<button onclick="changeContent()">Change Content</button>
```

```
<button onclick="changeStyle()">Change Style</button>
```

```
<button onclick="changeAttribute()">Change Attribute</button>
```

```
<button onclick="addElement()">Add Element</button>
```

```
<script>
```

```
function changeContent() {  
    document.getElementById("heading").innerHTML = "Welcome Students";  
    document.getElementById("message").innerHTML =  
        "Content changed using the DOM.";  
}
```

```
function changeStyle() {  
    document.getElementById("heading").style.color = "blue";  
    document.getElementById("heading").style.backgroundColor = "lightgray";  
    document.getElementById("message").style.fontSize = "20px";  
}
```

```
function changeAttribute() {  
    document.getElementById("image").src = "new.jpeg";  
    document.getElementById("image").width = 300;  
}
```

```
function addElement() {  
    let p = document.createElement("p");  
    p.innerHTML = "This paragraph was created using JavaScript.";  
    document.body.appendChild(p);  
}
```

```
</script>
```

```
</body>
```

```
</html>
```

7. CODE EXPLANATION

Code	Explanation
<code>document.getElementById("heading")</code>	Selects and retrieves the DOM element matching the specified ID.
<code>innerHTML</code>	Sets the inner HTML markup and visible text inside the selected element.
<code>element.style.color = "blue"</code>	Modifies the inline CSS font color property of the heading to blue.
<code>element.style.backgroundColor</code>	Applies a light gray background color to the heading element.
<code>element.style.fontSize = "20px"</code>	Sets the font size of the paragraph text to 20 pixels.
<code>image.src = "new.jpg"</code>	Updates the image resource path attribute dynamically.
<code>image.width = 300</code>	Modifies the rendered width attribute of the image element to 300px.
<code>document.createElement("p")</code>	Creates a new, empty paragraph element in memory.
<code>document.body.appendChild(p)</code>	Appends the newly created paragraph node to the end of the document body.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load the page in a browser	Default heading, paragraph, image (width 200), and 4 buttons displayed	Page rendered with default elements	Pass
TC-02	Click "Change Content"	Heading changes to "Welcome Students"; paragraph text updates	Content updated successfully	Pass
TC-03	Click "Change Style"	Heading text turns blue with gray background; font size 20px	Styles applied correctly	Pass
TC-04	Click "Change Attribute"	Image src becomes "new.jpg" and	Attributes updated	Pass

		width increases to 300	properly	
TC-05	Click "Add Element"	A new paragraph is dynamically appended to the page	Paragraph element appended	Pass

9. OUTPUT

Output 1: Initial Page Load

The page renders the default heading "Student Information", descriptive paragraph, image at 200px width, and four buttons.

Student Information

Click the button to change this content.



Change Content

Change Style

Change Attribute

Add Element

Output 2: After Clicking "Change Content"

The heading text is updated to "Welcome Students" and the paragraph text is updated to "Content changed using the DOM.".

Welcome Students

Content changed using the DOM.



Change Content

Change Style

Change Attribute

Add Element

Output 3: After Clicking "Change Style"

The heading text turns blue with a light gray background color, and the paragraph font size increases to 20px.

Welcome Students

Content changed using the DOM.



Change Content

Change Style

Change Attribute

Add Element

Output 4: After Clicking "Change Attribute"

The image source updates to "new.jpg" and the display width expands to 300px.

Welcome Students

Content changed using the DOM.



Change Content

Change Style

Change Attribute

Add Element

Output 5: After Clicking "Add Element"

A new paragraph stating "This paragraph was created using JavaScript." is appended to the bottom of the document.

Welcome Students

Content changed using the DOM.



Change Content

Change Style

Change Attribute

Add Element

This paragraph was created using JavaScript.

10. RESULT

Thus, a dynamic DOM manipulation web application was successfully designed and developed using JavaScript. The application demonstrates dynamic content modification, CSS style manipulation, attribute updating, and element creation and insertion. All five test cases passed successfully, confirming proper DOM behavior.